

WEIYU LI

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🌐 <https://github.com/wyysf-98>

EDUCATION

Hong Kong University of Science and Technology (HKUST)

Sept. 2023 – Now

Ph.D. candidate in ECE. Supervised by: Prof. Ping Tan

Interdisciplinary Research Center, Shandong University (IRC, SDU)

Sept. 2020 – Jun. 2023

M.Sc. in Computer Science and Technology. Supervised by: Prof. Baoquan Chen

Shandong University (SDU)

Sept. 2016 – Jun. 2020

Honors B.Eng. in Software Engineering. (overall ranked 4 out of 111)

EXPERIENCE

Meta AI

Research intern, working with Prof. Andrea Vedaldi on 3D Generation.

London, UK

StepFun

Research intern, working with Xuanyang Zhang on 3D Generation.

Shenzhen, China

Adobe

Research intern, working with Dr. Tuanfeng Y. Wang on Gaussian Splatting.

London, UK

Tencent AI-Lab

Research intern, working with Dr. Xuelin Chen on character motion generation.

Shenzhen, China

Visual Computing and Learning Lab, Peking University

Visiting student, worked on 3D neural scene generation.

Beijing, China

AICFVE, Beijing Film Academy

Intern(remote), worked on building digital human.

Qingdao, China

PUBLICATIONS

- [1] MeshFlow: Efficient Artistic Mesh Generation via MeshVAE and Flow-based Diffusion Transformer
Weiyu Li, Antoine Toisoul, Tom Monnier, Roman Shapovalov, Rakesh Ranjan, Ping Tan, Andrea Vedaldi
CVPR 2026 *highlight*
- [2] CraftsMan3D: High-Fidelity Mesh Generation with 3D Native Diffusion and Interactive Geometry Refiner
Weiyu Li*, Jiarui Liu*, Hongyu Yan*, Rui Chen, Yixun Liang, Xuelin Chen, Ping Tan, Xiaoxiao Long
CVPR 2025 *oral*
- [3] Step1X-3D: Towards High-Fidelity and Controllable Generation of Textured 3D Assets
Weiyu Li, Xuanyang Zhang, Zheng Sun, Di Qi, Hao Li, Wei Cheng, Weiwei Cai, Shihao Wu, Jiarui Liu, et al.
Tech Report, 2025
- [4] SweetDreamer: Aligning Geometric Priors in 2D Diffusion for Consistent Text-to-3D
Weiyu Li, Rui Chen, Xuelin Chen, Ping Tan
ICLR 2024
- [5] Example-based Motion Synthesis via Generative Motion Matching
Weiyu Li*, Xuelin Chen*, Peizhuo Li, Olga Sorkine-Hornung, Baoquan Chen
ACM Transactions on Graphics (SIGGRAPH), 2023
- [6] Patch-based 3D Natural Scene Generation from a Single Example
Weiyu Li*, Xuelin Chen*, Jue Wang, Baoquan Chen
Computer Vision and Pattern Recognition (CVPR), 2023
- [7] SceneMaker: Open-set 3D Scene Generation with Decoupled De-occlusion and Pose Estimation Model
Yukai Shi, **Weiyu Li**, Zihao Wang, Hongyang Li, Xingyu Chen, Ping Tan, Lei Zhang
CVPR 2026
- [8] Multi-View Large Reconstruction Model via Geometry-Aware Positional Encoding and Attention
Mengfei Li, Xiaoxiao Long, Yixun Liang, **Weiyu Li**, Yuan Liu, Peng Li, Wenhan Luo, Wenping Wang, Yike Guo
IEEE Transactions on Visualization and Computer Graphics (TVCG) 2025
- [9] UniTEX: Universal High Fidelity Generative Texturing for 3D Shapes
Yixun Liang, Kunming Luo, Xuelin Chen, Rui Chen, Hongyu Yan, **Weiyu Li**, Jiarui Liu, Ping Tan
CVPR 2026

- [10] PoseMaster: Generating 3D Characters in Arbitrary Poses from a Single Image
Hongyu Yan, Kunming Luo, **Weiyu Li**, Yixun Liang, Shengming Li, Jingwei Huang, Chunchao Guo, Ping Tan
CVPR, 2026
- [11] Dora: Sampling and Benchmarking for 3D Shape Variational Auto-Encoders
Rui Chen, Jianfeng Zhang, Yixun Liang, Guan Luo, **Weiyu Li**, Jiarui Liu, Xiu Li, Xiaoxiao Long, Jiashi Feng, Ping Tan
CVPR 2025
- [12] Iris3D: 3D Generation via Synchronized Diffusion Distillation
Yixun Liang, **Weiyu Li**, Rui Chen, Feipeng Tian, Jiarui Liu, Yuan-Chen Chen, Ping Tan, Xiaoxiao Long
ACM Transactions on Graphics 2025
- [13] BAGS: Building Animatable Gaussian Splatting from a Monocular Video with Diffusion Priors
Tingyang Zhang, Qingzhe Gao, **Weiyu Li**, Libin Liu, Baoquan Chen
arXiv preprint 2024
- [14] MoCo-Flow: Neural Motion Consensus Flow for Dynamic Humans in Stationary Monocular Cameras
Xuelin Chen, **Weiyu Li**, Daniel Cohen-Or, Niloy J. Mitra, Baoquan Chen
Computer Graphics Forum (Eurographics) 2022

SELECTED AWARDS & HONORS

- Two Special Prizes in RPA Parallel Computing Competition of Chinese Academy of Sciences (\$30, 000 bonus)
- The First Prize in the 12th Intel Cup Undergraduate Software Innovation Competition (\$2, 200 bonus)
- Two Second Prizes in Asian Student Supercomputer Challenge (ASC 2018 and 2019)
- The Second Prize in China Undergraduate Mathematical Contest in Modeling
- The First scholarship in Shandong University (Three times)
- Postgraduate Scholarship in The Hong Kong University of Science and Technology

PROFESSIONAL SERVICE

Conference Reviewer: ICLR, CVPR, NeurIPS, SIGGRAPH, SIGGRAPH Asia, CGI
Journal Reviewer: TVCG, ACM TOG, CVMJ

TEACHING EXPERIENCE

Teaching Assistant — ELEC 6910A First Principles of Computer Vision <i>Hong Kong University of Science and Technology</i>	2025 Spring
Teaching Assistant — ELEC 6910A First Principles of Computer Vision <i>Hong Kong University of Science and Technology</i>	2024 Fall
Teaching Assistant — ELEC 3120 Computer Networks <i>Hong Kong University of Science and Technology</i>	2024 Spring

TECHNICAL SKILLS

Programming Languages: Python, C++/C#, HTML/CSS, Java, JavaScript, SQL, L^AT_EX
Developer Tools: Pytorch, Tensorflow, OpenCV, Blender, three.js, node.js